

CreditMetrics AI - Technical Documentation

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Date: August 29, 2026

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1. Executive Summary

The CreditMetrics AI platform represents a complete, end-to-end Machine Learning pipeline designed for robust credit scoring and risk decisioning. The primary objective of this project was to automate the evaluation of loan applicants, minimizing default risk while maximizing approval accuracy using historical financial data.

By transitioning from legacy static systems to a highly interactive, API-driven web dashboard, the system now provides real-time model evaluation, live inference capabilities, and comprehensive audit trails.

2. Data Pipeline & Feature Engineering

The data pipeline was engineered to handle raw applicant data, processing it through rigorous validation and transformation stages.

- Missing Value Imputation: Handled skewed financial indicators using median imputation and categorical variables using mode substitution.
- Feature Engineering: Created synthetic features such as Debt-to-Income (DTI) ratio, Credit Utilization curves, and distinct credit rating buckets to enhance model separability.
- Encoding & Scaling: Applied One-Hot Encoding for categorical demographics and Standard Scaling for continuous variables to ensure gradient convergence.

3. Model Architecture & Training

Multiple algorithms were evaluated to find the optimal balance between interpretability and predictive power.

- Logistic Regression: Served as a high-interpretability baseline, providing clear weight coefficients for regulatory compliance.
- Decision Trees & Random Forests: Captured non-linear boundaries in borrower behavior. The Random Forest classifier significantly reduced overfitting through ensemble bagging.
- Hyperparameter Tuning: Utilized Grid Search with cross-validation to optimize tree depth, minimum samples per leaf, and class weights to handle dataset imbalances.

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4. Performance Metrics & Evaluation

The models were rigorously evaluated against industry-standard financial metrics.

- Achieved high ROC-AUC scores, ensuring the model effectively distinguishes between safe and risky borrowers.
- Prioritized Precision and Recall (PR-AUC) to minimize False Positives (approving bad loans) without severely restricting overall approval rates.
- Implemented live Confusion Matrix generation for threshold adjustments.

5. System Architecture & UI/UX

The backend is powered by a Flask REST API, strictly decoupling the Machine Learning logic from the presentation layer. The frontend is built utilizing TailwindCSS, delivering a 'Dark Mode' command-center aesthetic.

- Dynamic Endpoints: `/api/evaluate` and `/api/predict` serve JSON payloads directly to the frontend.
- Navigation Panel: Seamless routing between the Model Dashboard, Risk Assessment form, Data Pipeline viewer, and Audit Logs.
- Security: Immutable audit logging tracks all model execution queries for compliance.